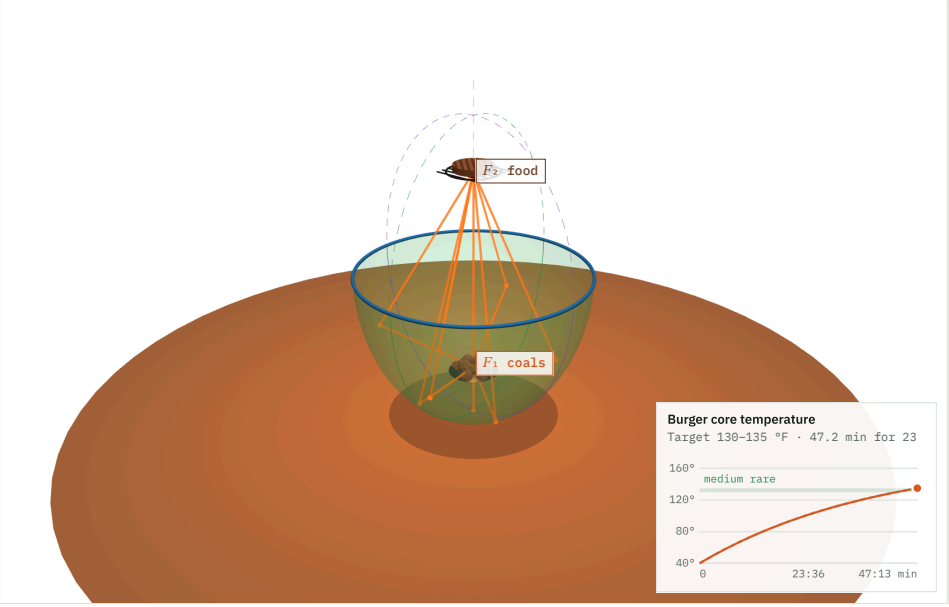
MATH 23 · AI-ASSIGNMENT 4 · QUADRIC SURFACES

The Ellipsoid Grill

Coals sit at focus F_1, the burger at focus F_2. Every heat ray that leaves the coals bounces off the wall and lands on the food. Change a and b, press Cook, and compare cooking time, capacity, and cost.



DONE Cooking time 47.2 min for 23 medium-rare burgers. Maximum on this grill: 23. Cost about $158.

Semi-axis a (height) 40.0 cm

Semi-axis b (radius) 30.0 cm

Kept below a so the two foci exist.

$c = \sqrt{(a^2-b^2)}$	$e = c/a$	$\eta = 0.4+0.5e$
26.46	0.661	0.731

Burgers to cook 23

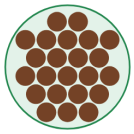
Full load (23 max)

COOK

Cooking time to medium rare
47.2 min
Cooked: 23 burgers to a 135 °F core

Max burgers: 23
Grill cost: $158

Shell surface area
1.389 m² (13,893 cm²)



Top view: 4-in (10.16 cm) patties in hexagonal packing on a disc of area πb^2.

▼ **Model assumptions**

T_o, reference time (min)	qty_o, reference load
6	4

Material price ($/m²)	Fixed hardware ($)
85	40

EQUATION & FOCI

$$z^2/a^2 + (x^2 + y^2)/b^2 = 1$$

$$z^2/1600 + (x^2 + y^2)/900 = 1$$

$c = \sqrt{(1600 - 900)} = 26.46$ cm. Coals at $F_1(0, 0, -26.46)$, food at $F_2(0, 0, +26.46)$. Focal distance $2c = 52.92$ cm.

COORDINATE-PLANE TRACES

- **z = 0** (xy-plane): $x^2 + y^2 = 900$, a circle of radius 30.0 cm. This is the rim.
- **x = 0** (zy-plane): $y^2/900 + z^2/1600 = 1$, an ellipse 60.0 cm wide and 80.0 cm tall.
- **y = 0** (xz-plane): $x^2/900 + z^2/1600 = 1$, the same ellipse turned 90°.
- Cross-section through a focus ($z = \pm c$) is a circle of radius $b^2/a = 22.50$ cm. Dashed arcs show the cut-away top half.

MODEL

$$S = 2\pi b^2[1 + (a / (b \cdot e)) \cdot \arcsin e]$$

$$T = T_0 \cdot (qty / qty_0) / \eta$$

$$Cost \approx S \cdot price + hardware$$

$S = 1.389$ m², $\eta = 0.731$, $T = 6 \cdot (23/4)/0.731 = 47.2$ min, cost = $1.389 \cdot \$85 + \$40 = \$158.09$.

Design comparison

Presets are examples. Each Cook adds your design. Click a row to load it. Green marks the best value in a column.

Clear my cooks

Design	a (cm)	b (cm)	c (cm)	e	η	S (m²)	Max burgers	Burgers	Time (min)	min / burger	Cost ($)	$ / burger slot
Preset · tall	50.0	30.0	40.00	0.800	0.800	1.658	23	23	43.1	1.88	181	7.87
Preset · default	40.0	30.0	26.46	0.661	0.731	1.389	23	23	47.2	2.05	158	6.87
Preset · round-ish	36.0	33.0	14.39	0.400	0.600	1.452	31	31	77.5	2.50	163	5.27
Preset · slim	50.0	20.0	45.83	0.917	0.858	1.046	8	8	14.0	**1.75**	**129**	16.11
Preset · large	60.0	45.0	39.69	0.661	0.731	3.126	**58**	58	119.1	2.05	306	**5.27**
My cook 1	40.0	30.0	26.46	0.661	0.731	1.389	23	23	47.2	2.05	158	6.87

Model only: medium rare is a 130–135 °F core. USDA advises 160 °F for ground beef. Heat-ray paths use the reflective property of the prolate spheroid: a ray from one focus reflects through the other.